

Inline Separation of Spotted Lanternflies (SLF)

Client Report

Team Di-Vine Intervention

JJ Bellinghier, Kayla Palumbo, Labhya Mehta, Mateo Garces, Suri Skomra

1. Context and Problem Statement

Mechanical grape harvesting operates at industrial scales of approximately 2–3 tons per hour but unintentionally collects Spotted Lanternflies (SLF) along with harvested grapes. Even low contamination levels (approximately 1–2 insects per 1000g) can result in rejected loads due to quality control and regulatory concerns.

Existing mitigation strategies such as manual sorting and post-harvest washing are not compatible with commercial throughput requirements. These approaches introduce significant processing delays, increase labor demand, and are incompatible with continuous harvesting operations.

1.1 Design objective

Develop a mechanical system for removing SLF during harvesting that preserves throughput and minimizes grape damage, and is compatible with both standalone operation and retrofit integration with existing conveyor-based harvesting systems.

2. Final Prototype and Application

A modular inline separation system was developed, integrating mechanical agitation and pneumatic separation to remove SLF during conveyor transport.

2.1 System Architecture

- Conveyor transport system for continuous material flow
- Directed airflow subsystem for insect separation
- Rotary brush subsystem for mechanical agitation of grape clusters

As grapes travel along the conveyor, they pass beneath a rotating brush that dislodges attached insects. A directed airflow system then separates SLF based on their lower mass and higher aerodynamic sensitivity relative to grapes. Grapes continue to the collection trough while the SLF are diverted into a separate containment bin.

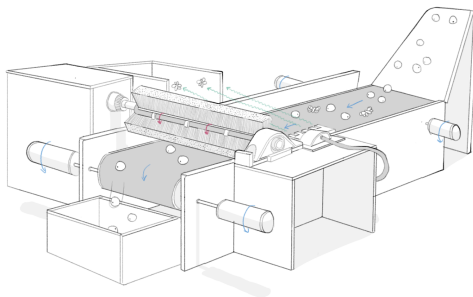


Figure 1: Conceptual design of full-scale inline SLF separation system.

This conceptual model represents the intended scalable deployment of the system in a commercial harvesting environment. The separation mechanism is maintained as a continuous flow process to ensure compatibility with high-throughput vineyard operations.

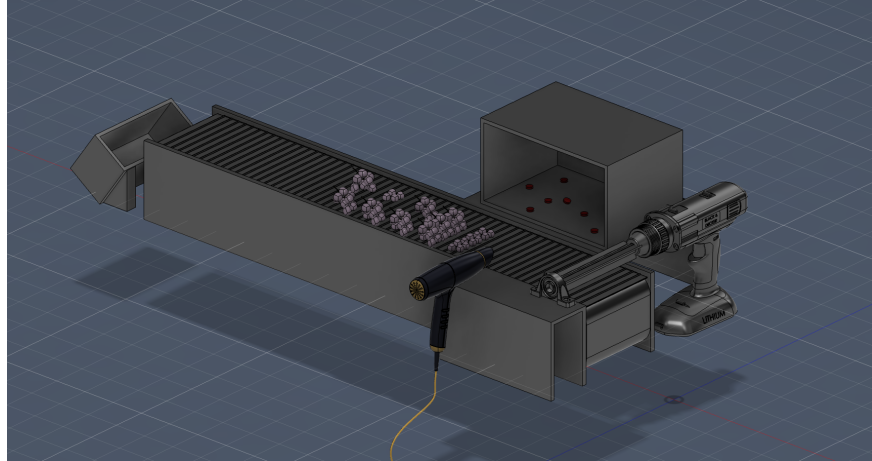


Figure 2: CAD model of proposed system implementation.

3. Conclusion and Recommendations

The prototype demonstrates that inline mechanical and pneumatic separation can effectively reduce SLF contamination without interrupting throughput or damaging grapes.

3.1 Recommendation

- Proceed with iterative development and scaled validation.

3.2 Recommended Next Steps:

- Improve brush uniformity to ensure consistent contact across full conveyor width
- Replace ad hoc airflow source with engineered compressed air manifold (target 5–10 m/s)
- Integrate vacuum-based capture to prevent re-entrainment of dislodged insects
- Conduct testing under higher throughput conditions representative of commercial harvesters

The system demonstrates sufficient performance to justify further development toward field-scale validation, with improvements focused on consistency and scalability.

4. Testing and Results

Experimental testing was conducted using fresh grapes and representative SLF models to evaluate separation performance and fruit integrity.

4.1 Key Performance Outcomes:

- Average SLF removal efficiency: 72%
- Observed variation in removal efficiency: 57–100% across conveyor width
- Optimal operating condition: ~500 RPM brush speed with high conveyor velocity and maximum airflow
- Grape integrity: no significant observable damage under tested conditions

Performance variation was primarily attributed to non-uniform brush contact across the conveyor width, indicating a need for improved mechanical consistency.

5. Prototype and Assembly Details

The prototype was constructed as a simplified conveyor-based system designed to replicate industrial harvesting flow.

5.1 Structural Frame

- Wooden base structure with supporting side walls
- Central alignment for conveyor integration

5.2 Conveyor System

- Cardboard rollers mounted on wooden dowels
- Paper-based belt reinforced with wax paper for durability
- Manual actuation via 3D-printed end knobs

5.3 Rotary Brush Subsystem

- Machined shaft driven by handheld drill
- Bristles mounted using custom 3D-printed shaft rings
- Epoxy bonding used to prevent relative motion and improve stability

5.4 Airflow Subsystem

- Hair dryer used as a surrogate for compressed air system during testing
- Positioned downstream of brush for immediate insect removal

5.5 Collection System

- Primary collection bin for grapes at conveyor exit
- Secondary side-mounted bin for SLF capture

5.6 Design Iterations

- Replaced geared conveyor concept due to jamming and misalignment
- Improved brush fixation to eliminate independent rotation
- Adjusted brush height and alignment to optimize contact consistency



Figure 3: Final prototype integrating conveyor, rotary brush, and airflow separation system.

6. Bill of Materials

Item	Quantity / Mass / Hour	Unit Cost	Cost Basis	Total Cost (\$)	Source
Brush	3	9.00	per unit	27.00	McMaster-Carr
Shaft	1	14.00	per unit	14.00	McMaster-Carr
Air Compressor	1	17.00	per unit	17.00	Amazon
Drill	1	79.99	per unit	79.99	Amazon
Adapter	1	9.99	per unit	9.99	Amazon
CAD (Mount)	25 g	0.05	per gram	1.25	RPL
CAD (Collection Trough)	786 g	0.05	per gram	39.30	RPL
CAD (Knobs)	20 g	0.05	per gram	1.00	RPL
Machining	3	5.00	per hour	15.00	MLS
Plywood	4	1.75	per unit	7.00	TDS
Plywood (alt)	3	3.00	per unit	9.00	TDS
Total Spent				220.53	

Table 1: Bill of Materials with unit-based and mass-based costing

6.1 Structural Components

- Wooden base boards (2) – TDS
- Wooden side walls (4) – TDS

6.2 Conveyor System

- Cardboard rollers (2) – TDS
- Wooden dowels (2) – TDS
- Paper conveyor belt with wax-paper coating – TDS
- 3D-printed drive knobs (4) – RPL

6.3 Brush System

- Machined steel shaft – McMaster-Carr
- Bristle material (brush segments) – McMaster-Carr
- 3D-printed shaft mounting rings – RPL
- Epoxy adhesive – TDS
- Handheld drill (drive source) – Amazon

6.4 Airflow System

- Hair dryer (temporary airflow generation unit) – Dyson (self-funded; not charged to project budget)

6.5 Collection System

- 3D-printed SLF collection bin – RPL
- Wooden mounting supports – TDS

6.6 Miscellaneous Materials

- Hot glue adhesive – TDS
- Fasteners and general mounting hardware – TDS
- Paper SLF test models – TDS
- Grapes (experimental validation material) – Mac's Cafe (self-funded; not charged to project budget))

7. References

- [1] evokeAG. “Harvest Optimisation Technology to Remove MOG.” 2021.
- [2] Kurtural, S. Kaan. “Mechanical Harvesting – Tools of the Trade.” UC Davis.
- [3] WECO Sorting. “The Science of Optical Sorting.” 2022.